

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC502 - MACHINE DESIGN - I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1	Which of the following is NOT a phase in the mechanical design process?			
	A) Conceptual design	B) Prototype testing	C) Mass production	D) Numerical control
	The purpose of using standardization and preferred numbers in design is to			
	A) Increase product uniqueness	B) Reduce variety and manufacturing cost	C) Complicate the selection process	D) Increase inventory size
3	The theory that is most suitable for ductile materials under static loading is			
	A) Maximum normal stress theory	B) Mohr's failure theory	C) Maximum distortion energy theory	D) Coulomb's failure theory
4	A knuckle joint is primarily used to connect two			
	A) Misaligned shafts	B) Plates under shear	C) Circular rods under tensile load	D) Gears for torque transmission
5	Which diagram is used for fatigue design under combined loading conditions?			
	A) Bode plot	B) Soderberg diagram	C) P-V diagram	D) Mohr's circle
6	The phenomenon of gradual failure due to repeated loading is called			
	A) Yielding	B) Buckling	C) Fatigue	D) Creep
7	ASME code recommends allowable shear stress for shaft design as			
	A) 100% of yield strength	B) 70% of ultimate strength	C) 30% of yield strength (for ductile materials)	D) Equal to bending stress

8	Which coupling is suitable for slight misalignment?			
	A) Rigid flange coupling	B) Muff coupling	C) Flexible coupling	D) Split muff coupling

9	A screw is said to be self-locking if			
	B) Lead angle < friction angle	B) Lead angle < friction angle	C) Efficiency > 90%	D) It can rotate under its own weight

10	In a welded joint, the throat area of a fillet weld is:			
	A) Equal to leg length	B) $0.707 \times \text{leg length}$	C) Double the leg length	D) Square root of leg length

11	The spring index (C) is defined as:			
	A) Wire diameter / Mean coil diameter	B) Mean coil diameter / Wire diameter	C) Number of coils / Wire diameter	D) Free length / Solid length

12	Shot peening increases spring life by			
	A) Increasing stiffness	B) Inducing compressive residual stress	C) Making it brittle	D) Increasing spring index

Q2. Solve the following.

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|----|---|---|
| A) | Explain the general industrial design procedure. | 6 |
| B) | Write a short note on standardization and preferred numbers in design | 6 |

Q3. Solve the following.

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|----|--|---|
| A) | Compare the various theories of failure under static loading. | 6 |
| B) | Design a knuckle joint to transmit a tensile load of 50 kN. The allowable stresses are Tensile stress = 75 Mpa, Shear stress = 60 Mpa, Crushing stress = 120 Mpa | 6 |

Q4. Solve Any Two of the following.

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|----|---|---|
| A) | Differentiate between Soderberg and Goodman criteria. | 6 |
| B) | A circular steel bar is subjected to a completely reversed axial load varying from +40 kN to -40 kN. The ultimate tensile strength of the material is 600 MPa, and the yield strength is 400 MPa. Use Soderberg's criterion to determine the diameter of the bar.
Take: Endurance limit = $0.5 \times \text{ultimate strength}$ = 300 Mpa & Factor of Safety (FOS) = 2 | 6 |
| C) | A machine component is subjected to a fluctuating bending load that varies from 30 kNm to 90 kNm. The material has: | 6 |

- Ultimate tensile strength = 700 MPa
- Yield strength = 500 MPa
- Endurance limit = 350 MPa
- Factor of safety = 2

Using Goodman's criterion, determine the diameter of a solid circular shaft required to withstand this load. Neglect stress concentration.

Q5. Solve Any Two of the following.

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|----|---|---|
| A) | Differentiate between rigid and flexible couplings. | 6 |
| B) | List the different types of keys with their uses. | 6 |
| C) | What are the ASME design considerations for shafts? | 6 |

Q6. Solve Any Two of the following.

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|----|--|---|
| A) | Explain self-locking and efficiency in power screws. | 6 |
| B) | Classify welded joints and explain stresses in a fillet weld. | 6 |
| C) | Design a helical compression spring to carry a static load of 1000 N with a deflection of 25 mm.
Given: | 6 |

- Spring index $C=6$
- Modulus of rigidity $G=80 \times 10^3 \text{ Mpa}$
- Maximum allowable shear stress $=400 \text{ MPa}$
- Use Wahl's correction factor

Find:

1. Wire diameter (d)
2. Mean coil diameter (D)
3. Number of active coils (n)

*** End ***